

that simplify supply chain processes and identify defects in fruits and vegetables, reducing wastage. “We also utilise shelf-monitoring cameras that feed into machine learning models to spot defects,” adds Sree Ram.

The shelf-monitoring solution includes an app called Johari, which uses computer vision models and Wi-Fi and IoT enabled cameras to identify pre-determined defects such as cuts and cracks in fresh produce. In the manual mode, an operator can submit a picture of the produce which is then analysed to highlight any item that does not meet quality standards. In the automated monitoring mode, cameras on shelves take pictures at predetermined intervals and assess for defects. The process takes six seconds.

Globally, Amazon has been among the early adopters of physical AI in its warehouses, operating the world's largest robotics fleet, according to the WEF report. Leveraging autonomous systems that bring together mobile robots, AI-based sorting and generative AI-guided manipulators to improve fulfilment has resulted in 25% faster delivery and boosted efficiency in some locations by 25%.

BUILD OR BUY

While several companies are going for customisation, some are adopting ready solutions such as autonomous mobile robots (AMRs). Mahindra & Mahindra is using physical AI with AMRs in production units and warehouses. For example, it has set up vision cameras which facilitate internal routing of sub-assemblies in production shops. “All AI initiatives we are working on in manufacturing are aimed at the shop floor. We have developed models to derive actions and are validating the accuracy of these actions. Once validated, these AI models will be integrated into our smart automation systems to scale up physical AI,” says Nalinikanth Gollagunta, CEO, Automotive Division, M&M. The company is working on a project where AI models will predict the integrity of a spot weld in real time. AMRs are a common sight on shop floors at ABB too. These are autonomous and embedded with a 3D camera, which detects obstructions, triggering a reroute if needed.

At the M&M manufacturing facilities, the smart automation-based inter-shop conveyors work by executing a predefined routing pattern with the PLC ensuring that the set route is always followed. Using physical AI enables flexible routing based on signals received from an AI model designed to find the shortest route and depending on current material movement across the lines.



AMIT CHADHA
CEO & MD, LTTS

“The biggest hurdle was not AI algorithms but unifying disparate data sources from decades-old systems”



JAYANTA BANERJEE
CIO, TATA STEEL

“Fundamentally, AI is not a standalone journey. You need to first invest in Cloud, Edge and data technology. You need to build the road before the car”

However, implementing the AI layer is not a straightforward process even if you have a robust foundation. At the ABB plant in Bengaluru, an AI-powered camera regulates turnstile access to a hardhat area where employees are required to be in personal protective equipment and safety shoes, something which wasn't always followed. The AI-enabled camera takes a picture and does a quick assessment of whether the person is wearing the appropriate gear; only then is he or she allowed in. The system now works efficiently and has even won an internal award, but Saju says it has taken them a lot of time to get to this stage. “Training the algorithm came with its own set of challenges. For instance, the system could not initially differentiate